

# Time as a Quantum Field: The Fundamental Unity of Temporal Dynamics and Physical Reality

Andriy Lemeshko

ORCID: 0000-0001-8003-3168

*Department of Software Engineering and Cybersecurity, State University of Trade and Economics, Kyiv, Ukraine*

## Abstract

This paper argues for a paradigm shift away from time as a passive coordinate, toward a model in which time is a **fundamental quantum field**  $\tau(\mathbf{x}, \Theta)$  that serves as the ontological foundation of physical reality. We show that unifying time and the quantum field into a single dynamical entity resolves several deep contradictions of contemporary physics:

- **Resolution of the “problem of time”:** the transition to a dynamical temporal field removes the “frozen universe” paradox of the Wheeler–DeWitt equation, since the evolution of the system becomes an intrinsic property of the field.
- **Ontological reduction:** Occam’s razor is applied to unify spacetime and matter (as topological solitonic excitations) into a single temporal condensate, closing the gap between geometry and quantum field theory.
- **Regularization of singularities:** natural quantization of the temporal field on the compact dimension  $S^1$  establishes a fundamental limit of divisibility, opening a mechanism for the natural regularization of physical singularities in black holes and in the early stages of the Big Bang.
- **The physical nature of the arrow of time:** thermodynamic asymmetry is interpreted as a relaxation process of the field potential  $V(\tau)$ , in which the growth of entropy is a dynamical consequence of the evolution of the temporal medium itself.

The paper contains a rigorous mathematical formalism (canonical quantization, 5D Kaluza–Klein reduction) and proposes concrete experimental verification criteria; in particular, it advances the LA-ODR hypothesis, which predicts a possible excitation frequency  $f_0 \approx 0.452$  MHz ( $\lambda_0 \approx 663$  m), subject to verification by high-performance phase-locking systems (TPLL) as an independent experimental criterion. The theory presented here is not merely a change of terminology but a fundamental paradigm shift that turns time into a measurable, quantized, and locally perturbable physical substance.

**Keywords:** temporal field, time quantization, chronons, Kaluza–Klein theory, emergent geometry, arrow of time, thermodynamic entropy, LA-ODR mechanism, TPLL, gravitational singularities, quantum gravity, compactification.

## Summary Paragraph

This paper elevates time from a passive coordinate to a fundamental quantum field  $\tau(\mathbf{x}, \Theta)$ . By treating time as a dynamical entity with its own degrees of freedom, we resolve the canonical “problem of time” in quantum gravity, regularize singularities through natural quantization, and

unify geometry and matter as emergent states of a single temporal condensate. The framework combines canonical quantization, Kaluza–Klein reduction, and information-geometric principles, while offering falsifiable predictions such as the LA-ODR resonance at  $f_0 \approx 0.452$  MHz. In this paradigm, the arrow of time is no longer an external assumption but a dynamical relaxation process of the temporal field itself.

## 1. Introduction: The Problem of Time in Modern Physics

The concept of time is at once the most universal and the least understood element of fundamental physics. Despite the unprecedented success of the two pillars of modern science — quantum mechanics (QM) and general relativity (GR) — their unification remains elusive, largely because of a fundamentally incompatible treatment of the nature of time. In each of the accepted physical paradigms, time plays a critical role, yet in none of them is it treated as a full-fledged physical object capable of possessing its own dynamics. This conceptual deficit leads to a series of deep theoretical crises.

### Historical Perspective

The search for a single underlying principle from which the diversity of physical phenomena can be derived is one of the oldest traditions in theoretical physics.

In different eras, different candidates for such a foundational substrate have been proposed. Nineteenth-century physics often viewed the ether as the fundamental medium underlying electromagnetic phenomena. Later, the development of field theory shifted attention toward electromagnetic and quantum fields as the primary constituents of reality. General Relativity introduced a different perspective, identifying the geometry of spacetime itself as the fundamental structure governing gravitational phenomena.

The present work follows the same methodological tradition, without presupposing that its outcome will be comparable in scope or success. It does not introduce a new scientific method, but rather explores a different candidate for the role of the primary substrate.

The central hypothesis of the Temporal Theory of the Universe (TTU) is that time is not merely an external parameter used to describe physical processes. Instead, it is treated as a physical field possessing its own dynamics. Within this framework, geometry, gravity, matter, and information processes are investigated as possible emergent manifestations of a deeper temporal structure.

Whether this hypothesis is ultimately correct remains a matter for experiment and mathematical consistency. The purpose of the present work is therefore not to claim a final description of nature, but to explore the consequences of adopting time as the primary dynamical entity and to examine whether known physical structures can be consistently reconstructed from that assumption.

**Terminological standard.** To avoid the conceptual confusion that has accompanied discussions of the nature of time for centuries, we introduce a strict terminological distinction that will be maintained throughout this work:

- The symbol  $t$  (or  $t_{\text{obs}}$ ) denotes exclusively the **macroscopic observable quantity** — the coordinate measured by clocks. It is a *derived*, emergent property arising as the vacuum expectation value of an underlying dynamics.

- The symbol  $\tau(x, \Theta)$  denotes the **fundamental temporal field** — the primary dynamical object, a quantum field with its own degrees of freedom, Lagrangian, and equation of motion.

The ontological relation between them is fundamental: macroscopic time  $t$  arises as the condensate (vacuum expectation value) of the temporal field:

$$t_{\text{obs}} \sim \langle \tau(x, \Theta) \rangle, \quad (1)$$

where the averaging is performed over quantum fluctuations and over the compact dimension  $\Theta$ . Thus, in this paper we do not “quantize time”; we elevate time to the status of a field, while the observable time  $t$  remains the classical limit of that field.

### 1.1. Time in Classical Physics: A Coordinate Without Dynamics

In classical Newtonian mechanics, time  $t$  is postulated as an absolute, true, and mathematical entity that “flows uniformly, without relation to anything external.” In this formalism, time acts as an ideal background metronome.

Mathematically, classical time is merely a parameter of evolution. It has no equation of motion of its own, possesses no energy or momentum, does not interact with matter, and experiences no back-reaction from physical processes. Here time is an absolutely rigid, passive stage on which the dynamics of physical bodies unfold. In classical physics, time has no dynamics — it simply sets the rhythm.

### 1.2. Time in Quantum Mechanics: An External Parameter That Is Not Quantized

In standard quantum mechanics, despite its radical departure from classical determinism, the treatment of time remains strikingly Newtonian. In the Schrödinger equation:

$$i\hbar \partial\psi/\partial t = H\psi \quad (2)$$

time  $t$  appears as a continuous, external evolution parameter.

The fundamental asymmetry of quantum theory lies in the fact that position (the coordinate  $x$ ) and momentum ( $p$ ) are elevated to the status of Hermitian operators  $\hat{x}$  and  $\hat{p}$ , which obey Heisenberg’s uncertainty principle. Time  $t$ , by contrast, is not an operator (the well-known Pauli theorem states that introducing a self-adjoint time operator canonically conjugate to the Hamiltonian leads to an energy spectrum unbounded from below). Thus quantum mechanics quantizes everything except time itself. Time remains an unquantized, classical parameter that exists, as it were, “outside” the quantum system, dictating its unitary evolution.

### 1.3. Time in General Relativity: A Geometric Coordinate Fused with Space

Albert Einstein, in special and later in general relativity, stripped time of its absolute status, turning it into the fourth dimension of a pseudo-Riemannian manifold — spacetime. Time became local and relative: it slows in gravitational fields and at speeds approaching that of light.

Nevertheless, despite this geometric flexibility, time in GR ( $x^0$ ) remains merely a coordinate. Together with the spatial dimensions, it forms the metric tensor  $g_{\mu\nu}$ , which describes the gravitational field. But the coordinate  $t$  itself has no physical substance of its own; it is dissolved

into geometry. In Einstein's "block universe" picture, past, present, and future exist simultaneously as a static four-dimensional geometry. Time does not "flow" — it is simply a measure of the distance between events along a worldline.

#### **1.4. Consequences: The Problem of Time, the Absence of a Unified Description, and the Black-Hole Information Paradox**

The attempt to unify quantum mechanics (in which time is an external, absolute parameter of evolution) with general relativity (in which time is a local, dynamically curved coordinate) leads to catastrophic contradictions:

- **The Problem of Time:** canonical quantization of gravity yields the Wheeler–DeWitt equation,  $\hat{H}\Psi = 0$ . This equation, which describes the wave function of the entire universe, contains no time derivative. The global universe turns out to be static; time mathematically vanishes from the fundamental description, in contradiction with our macroscopic experience of change.
- **The black-hole information paradox:** according to GR, gravitational collapse into a black hole produces a singularity where the curvature of spacetime diverges and time "stops." When the black hole evaporates via Hawking radiation, a conflict arises: if time as a coordinate ends at the singularity, where does the quantum information go? The absence of a unified temporal background makes a correct description of this process impossible.
- **Singularities:** the infinities of the Big Bang and of black holes indicate that the classical continuous model of time has exhausted itself. The continuity of the coordinate  $t$  allows energy to be compressed into an infinitesimally small point.

Singularities and information paradoxes are a consequence of treating time as the "stage" and matter fields as the "actors." However, if we recognize that time and the quantum field are one and the same physical entity, the mathematical infinities disappear. In this paper we show that time is not a background; it is the primary quantum field, whose excitations give rise to all known physical phenomena.

**Conclusion to Section 1:** all of these unresolved paradoxes point to a fundamental error — treating time exclusively as a passive coordinate (GR) or a background parameter (QM) is mistaken. The current crisis in theoretical physics demands a radical revision of ontology: time must be endowed with its own physical degrees of freedom, integrated into the structure of a quantum field.

## **2. Why Unifying Time and the Field Is Logical: Four Arguments**

To overcome the crisis described in the previous section, one must abandon the notion of time as an abstract mathematical background. The Temporal Theory of the Universe (TTU) proposes a radical but logically inevitable step: time must be elevated to the status of a fundamental quantum field  $\tau(x, \Theta)$ .

In this paradigm, time ceases to be merely the coordinate in which events occur and becomes a physical substance possessing its own energy, dynamics, and potential. This unification rests on four deep physical arguments that resolve key contradictions of contemporary theoretical physics.

## 2.1. Argument 1: Eliminating the “Frozen Time” Paradox in Quantum Gravity

As noted above, canonical quantization of general relativity leads to the Wheeler–DeWitt equation ( $\hat{H}\Psi = 0$ ), in which the evolution of the system is absent because the Hamiltonian is a constraint equal to zero.

If, instead, we recognize time not as the coordinate  $t$  but as the dynamical field  $\tau$ , the “frozen universe” paradox disappears. The field  $\tau$  possesses its own physical degrees of freedom and a conjugate momentum. In this formulation, the evolution of the universe is measured not relative to an external abstract clock but relative to the internal states of the temporal field itself. Changes in physical quantities correlate with changes in the amplitude and phase of the field  $\tau$ . Dynamics returns to quantum gravity because time becomes a material referent of evolution.

## 2.2. Argument 2: Ontological Reduction (Occam’s Razor)

Modern physics suffers from an overly complex basic ontology, in which an “arena” (spacetime) is separated from “actors” (quantum matter fields). The principle of emergence, together with Occam’s razor, demands reducing these entities to a single common denominator. Temporal theory proposes an elegant reduction: there exists only one fundamental field — the temporal condensate  $\tau(x, \Theta)$ . In this picture:

- **Space and gravity** arise as a macroscopic manifestation of the phase coherence and spatial gradients of the field  $\tau$ .
- **Elementary particles (matter)** are interpreted as localized topological vortices (solitons) of the same temporal medium.

Thus matter and space appear not as two independent entities but as different structural states of a single temporal field. This conceptually unifies geometry (GR) and quantum field theory (QFT), removing the ontological dualism that for centuries has been an obstacle to building a complete theory of quantum gravity.

## 2.3. Argument 3: Natural Regularization of Singularities Through Quantization of the Temporal Field

Singularities at the centers of black holes and at the moment of the Big Bang are a sign that the classical concept of a continuous spacetime breaks down.

However, any fundamental field in physics obeys the laws of quantum mechanics, and hence its excitations are discrete. If time is the field  $\tau(x, \Theta)$ , it must possess a discrete spectrum of states at the fundamental (Planck) scale. The quanta of this field — chronons — establish an absolute limit of divisibility for any physical process.

A physical system cannot be compressed to a state of infinite density, since it cannot exceed the minimal quantum of phase volume of the temporal field. Thus, turning time into a quantum field opens a natural mechanism for regularizing gravitational singularities, one that requires further numerical investigation within specific collapse models.

## 2.4. Argument 4: The Thermodynamic Arrow of Time as Dynamics of the Field Potential

The microscopic laws of physics are symmetric under time reversal ( $t \rightarrow -t$ ). Yet the macroscopic universe exhibits a definite “arrow of time,” associated with the growth of entropy.

If time is a physical field, it must have its own dynamics, governed by an energy functional (the potential  $V(\tau)$ ). In this paradigm, the “flow of time” ceases to be an abstraction and becomes a physical process of relaxation of the temporal field toward its vacuum minimum.

The asymmetry of time (the thermodynamic arrow) becomes a consequence of the fundamental nonlinear dynamics of the field  $\tau$ . Entropy increases not merely through statistical probability, but because the temporal field physically “rolls down” its potential, creating the direction of macroscopic causality.

**Conclusion to Section 2:** granting time the status of a fundamental quantum field is not a linguistic game but a necessary architectural step that resolves the Wheeler–DeWitt problem, eliminates singularities, and reduces spacetime and matter to a single temporal denominator.

## 3. Mathematical Formalism: The Temporal Field $\tau(x, \Theta)$

The transition from time as an abstract parameter  $t$  to the quantum field  $\tau(x, \Theta)$  is not merely a change of variables but a fundamental change in the physical status of time. The fundamental dynamical object is now the field  $\tau$ , which has its own Lagrangian, equation of motion, and quantum excitations, while the observable time  $t_{\text{obs}}$  arises as its vacuum expectation value. We will show that this unification allows gravity and electromagnetism to be derived as effective modal states of this field.

To turn the philosophical concept of dynamical time into a rigorous physical apparatus, we construct a canonical theory of the temporal field, drawing on the methods of differential geometry and quantum field theory. The principal task is to replace the passive evolution parameter  $t$  with the active scalar field  $\tau$ .

### 3.1. The Field as Carrier of Time: $\tau(x, \Theta)$ in Place of the Coordinate $t$

In the Temporal Theory of the Universe (TTU), the fundamental object is a physical scalar field  $\tau$  distributed over a five-dimensional manifold. The observable macroscopic time  $t_{\text{obs}}$  arises as the vacuum expectation value of this field:

$$t_{\text{obs}} \sim \langle \tau(x, \Theta) \rangle, \quad (3)$$

where the averaging is performed over quantum fluctuations and over the compact dimension  $\Theta$ . Physical processes evolve not relative to an external clock but relative to the local state of the temporal condensate. From now on we deliberately avoid identifying  $\tau$  with  $t$ ; these are distinct ontological levels.

### 3.2. The Hyper-Temporal Axis $\Theta$ : A Euclidean Thermal Circle as the Internal Coordinate of Quantum Evolution

If time  $t$  is replaced by the field  $\tau$ , a critical methodological question arises: relative to what does this field itself evolve? What serves as its independent argument? A naive answer leads to infinite regress (one would need “a clock for the clock”).

To avoid this regress, we introduce the extended manifold  $M_5 = M_4 \times S^1$ , where  $S^1$  is a compact dimension. We denote the coordinate on this dimension  $\Theta$  (hyper-time, or the phase component). It is important to emphasize, however, that  $\Theta$  is not a “flow of time” in the usual sense — it is not a dynamical axis along which something flows.

Within TTU,  $\Theta$  is interpreted as a Euclidean thermal coordinate (a thermal circle), in close analogy with finite-temperature field theory [6]. On this reading:

- The period of the compact dimension is  $2\pi R_\Theta$ , corresponding to the inverse temperature of the temporal condensate:

$$\beta_\tau = 2\pi R_\Theta, \quad T_\tau = 1/(2\pi R_\Theta) \quad (4)$$

(in Planck units).

- Evolution along  $\Theta$  does not describe a macroscopic “flow” but the **spectral structure** of quantum modes (analogous to a Fourier series or to discrete energy levels in a heat bath). This is why the field is a function  $\tau = \tau(x^\mu, \Theta)$  —  $\Theta$  acts as an internal spectral parameter, not an external metronome.

#### Why this removes the regress (Clausius vs. Klein):

This interpretation radically changes the ontological status of  $\Theta$ :

- **In the Klein sense (geometry):**  $\Theta$  is the fifth dimension, whose reduction yields a gauge field  $A_\mu$  and a tower of massive modes. This supplies the mathematical apparatus.
- **In the Clausius sense (thermodynamics):**  $\Theta$  is a thermal (Euclidean) coordinate whose period encodes the entropic potential of the condensate. The arrow of time (entropy) is generated not by motion along  $\Theta$  but by relaxation of the potential  $V(\tau)$  (see Section 6).

Thus, the field  $\tau$  does not evolve “relative to”  $\Theta$  as a clock; rather, it is “decomposed” into spectral components along  $\Theta$ .  $\Theta$  plays the role of a phase space of states, while the macroscopic time  $t_{\text{obs}}$  arises as a correlation of the zero mode ( $n = 0$ ) with the thermodynamic gradient. This entirely removes the logical regress.

*Note on interpretive scope.* The thermal (Matsubara-type) reading of  $\Theta$  is used here only as an interpretational analogy, motivating the entropic role of the potential  $V(\tau)$  discussed in Section 6. The compact coordinate  $\Theta$  is introduced geometrically through the compactification of an additional dimension on  $S^1$ ; its thermodynamic reading should be understood as heuristic rather than as a strict identification with a finite-temperature field theory.

### 3.3. Canonical Structure: $[\tau, p_\Theta] = i\hbar$

The field  $\tau$  has a conjugate momentum:

$$p_\Theta = \partial L / \partial (\partial_\Theta \tau). \quad (5)$$

Since  $\Theta$  is a compact spectral coordinate (and not a coordinate of external evolution, as established above in §3.2), canonical quantization should be carried out separately for the spatial field part and for the discrete  $\Theta$ -mode, in order to avoid mixing the delta function over  $x$  with the operator on the compact dimension. For the canonical momentum conjugate to  $\tau$  with respect to the spatial coordinates, the standard field commutation relation is introduced:

$$[\tau(x, \Theta), \pi_\tau(x', \Theta')] = i\hbar \delta^3(x - x') \delta(\Theta - \Theta'), \quad (6)$$

where  $\pi_\tau \equiv \partial L / \partial (\partial_{\text{obs}} \tau)$  is the ordinary canonical momentum conjugate to  $\tau$  with respect to macroscopic time. For the compact coordinate  $\Theta$  itself, the operator  $\hat{p}_\Theta$  acts not as a continuous momentum but as a modal (Kaluza–Klein) operator:

$$\hat{p}_\Theta = -i\hbar \partial / \partial \Theta, \quad \hat{p}_\Theta \psi_n = (\hbar n / R_\Theta) \psi_n. \quad (7)$$

This equation is the mathematical heart of the theory. Quantum fluctuations of the field  $\tau$  become an integral part of physical reality, naturally replacing the artificial introduction of time uncertainty found in standard quantum mechanics.

### 3.4. Lagrangian and Equation of Motion

$$L = (1/2) g^{MN} (\partial_M \tau) (\partial_N \tau) - V(\tau), \quad (8)$$

where the indices  $M, N$  range from 0 to 4 (four spacetime indices plus  $\Theta$ ).

In particular, the  $\Theta\Theta$ -component of the metric gives a separate term responsible for hyper-temporal stiffness:  $g_{\Theta\Theta} \equiv \kappa$ , so that the term  $(1/2)\kappa(\partial_\Theta \tau)^2$  is already implicitly contained in (8) as part of the full sum over  $M, N$ ; the explicit form of the hyper-temporal term is consistent with the canonical TTU-5D action ( $L_\tau = (1/2)\alpha(\nabla \tau)^2 - \beta\tau^2 + (1/2)\kappa(\partial_\Theta \tau)^2$ ).

The principle of least action leads to the classical equation of motion (the analogue of the Klein–Gordon equation with a nonlinear source):

$$\square_5 \tau + \partial V / \partial \tau = 0, \quad (9)$$

where  $\square_5$  is the five-dimensional d'Alembert operator. This equation describes how perturbations of the temporal field propagate and relax under the action of the potential  $V(\tau)$ .

### 3.5. Linearization and Effective Mass: $m_{\text{eff}}^2 = V''(\tau_0)$

Considering small fluctuations  $\delta\tau = \tau - \tau_0$  near the minimum of the potential  $V(\tau)$ , we expand it in a Taylor series. Since  $V'(\tau_0) = 0$ , the second derivative determines the mass of the excitations:

$$m_{\text{eff}}^2 = \partial^2 V / \partial \tau^2 |_{\tau=\tau_0}. \quad (10)$$

The mass  $m_{\text{eff}}$  determines the “stiffness” of the condensate and the characteristic radius of temporal correlations. Excitations of this field — hypothetical quanta of the temporal field (scalar chronons) — have an effective mass  $m_{\text{eff}}$ . The presence of the nonlinear potential  $V(\tau)$  allows the field to form stable topological structures (solitons), which constitute the fundamental building blocks of matter.

**Conclusion to Section 3:** the mathematical formalism developed in this section demonstrates that the transition from time as a background coordinate  $t$  to the quantum temporal field  $\tau(x, \Theta)$  is not only logically justified but also mathematically rigorous.

## 4. How the Field Generates Geometry and Matter

If time is a fundamental quantum field  $\tau(x, \Theta)$ , a logical question arises: where do the familiar spacetime and matter come from? The Temporal Theory of the Universe (TTU) proposes a radically simplified ontology: both geometry and matter are emergent (derived) phenomena. They are macroscopic consequences of the dynamics of a single temporal condensate. In this section we show how perturbations of the time field mathematically generate the metric, curvature, gravity, and matter itself.

### 4.1. The Metric as a Functional of $\tau$ : $g_{\mu\nu} = (\tau/\tau_0) \eta_{\mu\nu}$

In general relativity, the metric tensor  $g_{\mu\nu}$  is the fundamental variable that defines geometry. In our paradigm, the metric is not fundamental. It arises as a macroscopic functional of the local density of the temporal field.

In the simplest approximation, the effective spacetime metric is formed as a conformal map of the underlying (flat) space  $\eta_{\mu\nu}$ , with the normalized time field serving as the conformal factor:

$$g_{\mu\nu} = (\tau/\tau_0) \eta_{\mu\nu}, \quad (11)$$

where  $\tau_0$  is the background (vacuum) value of the temporal condensate. The physical meaning of this equation is profound: the local “stretching” or “compression” of space is determined solely by how dense the temporal field is at that point. Where the field  $\tau$  deviates from the vacuum value  $\tau_0$ , spacetime acquires effective curvature. Space ceases to be an independent entity — it becomes a macroscopic manifestation of the phase coherence of temporal modes.

### 4.2. Curvature as a Derivative of the Field

If the metric is a function of the time field, then the scalar Ricci curvature ( $R$ ), which in GR describes the geometric curvature of space, must reduce mathematically to derivatives of this field.

Substituting the conformal metric from the previous subsection into the standard definition of the Ricci tensor (the full step-by-step derivation is given in Appendix D), we obtain the scalar curvature as a direct consequence of the kinematics of the temporal field:

$$R[\tau] = (\tau_0/\tau) [-3 \square \tau / \tau + (3/2)(\partial\tau/\tau)^2], \quad (12)$$

where  $\square$  is the d’Alembert (wave) operator acting on the field  $\tau$ , and  $(\partial\tau)^2$  is the square of its four-gradient. In this expression, the operator  $\square$  and the gradient  $\partial$  are evaluated with respect to the flat background metric  $\eta_{\mu\nu}$ , which is the standard convention for conformal transformations. This equation demonstrates that what Einstein called “the curvature of spacetime” is not the abstract geometry of empty space but a real hydrodynamic stress (gradients and wave-like perturbations) of the temporal medium. The curvature of space is the dynamics of time written in geometric language.

### 4.3. Matter as Solitonic Excitations: Topological Stability and the Three-Generation Bound

Where does matter come from if, at the fundamental level, only the time field exists? According to the principles of nonlinear field theory, matter is not a foreign object placed into space. Elementary particles (fermions) are interpreted as stable, localized topological vortices (solitons) of the temporal field itself. The mass of a particle in this picture is the localization energy of the vortex, required to hold its shape against the elasticity of the temporal medium.

Moreover, this model offers a natural explanation of a puzzle of the Standard Model: why exactly three generations of elementary particles exist in nature (for example, the electron, the muon, and the tau lepton). In the nonlinear theory of the temporal field, the stability of a chronon knot is governed by a topological winding index  $n \in \mathbb{N}$ . The energy of an  $n$ -vortex configuration of a chronon knot on the compact dimension  $S^1$  is given by the volume functional:

$$E_n = \int d^3x \int_{S^1} d\Theta [\alpha(\nabla\tau_n)^2 + \kappa(\partial_\Theta\tau_n)^2 + V_{\text{sat}}(|\nabla\tau_n|)], \quad (13)$$

where  $\alpha$  is the temporal stiffness,  $\kappa$  is the hyper-temporal inertia, and  $V_{\text{sat}}$  is a nonlinear Born–Infeld-type saturation potential that prevents unbounded growth of the gradient. The hyper-temporal contribution scales as  $(\partial_\Theta\tau)^2 \sim (n\tau_0/R_\Theta)^2$ , meaning that higher windings inevitably increase the local temporal stress.

A chronon knot is dynamically stable only as long as the saturation regime is nowhere reached:

$$|\nabla\tau| < \tau_{\text{crit}}. \quad (14)$$

The minimal temporal gradient required for an  $n$ -vortex configuration scales as  $|\nabla\tau|_{\text{min}} \approx (n\tau_0)/(\lambda_C R_\Theta)$ , where  $\lambda_C$  is the characteristic Compton-like localization scale of the relevant family of modes. This yields a strict upper bound:

$$n < \tau_{\text{crit}} \lambda_C R_\Theta / \tau_0 \Rightarrow n_{\text{max}} \approx 3. \quad (15)$$

For  $n \geq 4$ , the required phase energy and gradients inevitably drive the soliton into the global saturation regime  $|\nabla\tau| \rightarrow \tau_{\text{crit}}$ . In this regime the restoring force vanishes, the knot loses stability, and the configuration decays via emission of scalar excitations  $\delta\tau$  into the background condensate. The estimate is robust to order-of-magnitude variations in  $\lambda_C$  and  $R_\Theta$ , and its mechanism — nonlinear saturation of the field gradient — introduces no additional free parameters, in contrast to phenomenological fitting models.

Thus, the proposed model offers a natural explanation for the bound of three generations of matter, relating it to the geometric criterion of temporal-gradient saturation. Each generation corresponds to a stable topological charge  $n = 1, 2, 3$ , while configurations with  $n \geq 4$  inevitably enter the saturation regime  $|\nabla\tau| \rightarrow \tau_{\text{crit}}$  and lose stability. Unlike phenomenological models with free gauge parameters, the bound  $n_{\text{max}} \approx 3$  is expressed directly through the geometric parameters of the temporal medium —  $\tau_{\text{crit}}$ ,  $\lambda_C$ ,  $R_\Theta$ , and  $\tau_0$  — without additional fitting coefficients. Numerical verification of this result for specific parameter values remains a task for future research.

#### 4.4. Gravity as a Logarithmic Potential of the Temporal Field: $\Phi = c^2 \ln(\tau/\tau_0)$

Since the curvature of space is reduced to the dynamics of the field  $\tau$ , the gravitational interaction also receives a new, temporal explanation. The canonical form of the temporal gravitational potential at the core of TTU (TTG-1) is determined not by the square of the gradient of the field itself, but by a logarithmic potential:

$$\Phi(x) = c^2 \ln(\tau(x)/\tau_0). \quad (16)$$

In the weak-field limit ( $|\Phi| \ll c^2$ ), this reduces to the standard Newtonian approximation  $\ln \tau \approx \Phi/c^2$ . The quantity that drives the motion of a test body is not the potential itself but the gradient of the logarithm of the field, multiplied by a coefficient of temporal susceptibility of matter  $\chi_\tau$ :

$$a = -\chi_\tau \cdot c^2 \cdot \nabla \ln \tau. \quad (17)$$

Here  $\chi_\tau$  is the dimensionless temporal susceptibility of matter (for ordinary, incoherent matter,  $\chi_\tau = 1$ , and the expression reduces to the standard Newtonian acceleration). It is essential that the driver of motion is the gradient of the **logarithm** of the field ( $\nabla \ln \tau$ ), not the gradient of the field itself ( $\nabla \tau$ ): structure and dynamics are separated — the temporal geometry ( $\nabla \ln \tau$ ) sets the direction of admissible motion, while  $\chi_\tau$  determines whether, and to what extent, a given form of matter responds to that gradient. In this paradigm, massive bodies create a local logarithmic deficit (or excess) of temporal density around themselves; other objects “fall” toward the massive body not because they are pulled by an invisible force, but because they move along the gradient of the logarithm of the temporal field. Gravity is a structural-dynamical response of matter to the geometry of time, rather than a direct action of the square of the field gradient.

**Conclusion to this section:** this section is the core of the ontological reduction. Using a rigorous mathematical relation among the metric, curvature, and the scalar field  $\tau$ , we remove the need for the independent existence of space and matter. Everything reduces to a single entity: localized perturbations of the field  $\tau$  give rise to particles, while its gradients create geometry and gravity.

### 5. Quantization of the Temporal Field: Hypothetical Quanta of the Temporal Field and Their Spectrum

If time is a physical field, it inevitably obeys the laws of quantum mechanics. In the classical paradigm, the division of time into ever-smaller intervals could continue indefinitely. In the Temporal Theory of the Universe (TTU), however, the presence of a compact hyper-temporal dimension  $\Theta$  imposes strict topological constraints, leading to a discrete spectrum of excitations. In this section we show how hypothetical quanta of the temporal field arise from the continuous field, and we distinguish two types of such excitations according to their spin and physical role.

#### 5.1. Discrete Modes in $\Theta$ : $\tau(x, \Theta) = \sum_n \tau_n(x) e^{in\Theta/R_\Theta}$

Since the additional phase dimension  $\Theta$  is compact (it has the topology of a circle  $S^1$  with radius  $R_\Theta$ ), any field on it must be periodic. This allows the temporal field to be expanded in a Fourier series:

$$\tau(x, \Theta) = \sum_{n=-\infty}^{\infty} \tau_n(x) e^{in\Theta/R_\Theta}, \quad (18)$$

where:

- $n \in \mathbb{Z}$  is the harmonic (mode) number;
- $\tau_n(x)$  are the effective four-dimensional fields;
- $R_\Theta$  is the compactification radius.

The zero mode ( $n = 0$ ) is independent of  $\Theta$  and accounts for the macroscopic, continuous flow of time. Higher harmonics ( $n \neq 0$ ) represent discrete perturbations of the temporal condensate.

## 5.2. Energy Spectrum: $E_n = \hbar\omega_n = \hbar\sqrt{[\kappa(n/R_\Theta)^2 + (m_\tau c/\hbar)^2]}$

From the equation of motion for the 5D scalar field, a dispersion relation follows:

$$E_n = \hbar\omega_n = \hbar\sqrt{[\kappa(n/R_\Theta)^2 + (m_\tau c/\hbar)^2]}, \quad (19)$$

where  $\kappa$  is the coupling constant of temporal transport and  $m_\tau$  is the effective mass (determined by the curvature of the potential  $V''(\tau_0)$ ). The excitation energy is quantized; at low energies ( $E \ll \hbar/R_\Theta$ ), we interact only with the zero mode, so that time appears classical and continuous. Under extreme conditions (black hole centers, the Big Bang), higher modes become active.

## 5.3. Two Types of Excitations: Scalar Chronons and Vector Gauge Bosons

In earlier versions of this work, the quanta of the temporal field were often simply called “chronons,” which led to terminological imprecision. In fact, it is necessary to distinguish two fundamentally different types of excitations of the temporal medium, which have different spins and play different physical roles:

- **Scalar chronon-dilatons (spin 0).** These are direct excitations of the scalar field,  $\delta\tau_n(x) = \tau_n(x) - \tau_0$ . They arise from fluctuations in the density of the temporal condensate and have a mass  $m_\tau$ , determined by the potential  $V(\tau)$ . These excitations are responsible for oscillations in the density of time and may be linked to inflationary or dark-energy dynamics. Their interaction with matter is extremely weak, which explains their “invisibility” in present-day experiments.
- **Vector gauge bosons (spin 1).** These quanta do not originate from the scalar field  $\tau$  but from the geometry of the compact dimension. As shown in Section 4 (and in the standard Kaluza–Klein reduction [1, 2]), the off-diagonal components of the 5D metric,  $g_{\mu\Theta}(x)$ , behave as a vector field  $A_\mu(x)$  with  $U(1)$  gauge symmetry upon reduction on  $S^1$ . These vector modes are the carriers of the electromagnetic interaction, and their quanta are photons. Electromagnetism thus arises not from the field  $\tau$ , but from the nontrivial topology of spacetime.

The term “chronon” should therefore be applied exclusively to the scalar excitations  $\delta\tau_n$ , while the vector modes  $A_\mu$  are ordinary gauge bosons whose origin is geometric rather than field-theoretic in the sense of  $\tau$ . This distinction removes the inconsistency in which the scalar field was assigned the role of a force carrier — a role proper only to vector fields.

**Conclusion to this section:** quantization of the temporal field on a compact dimension naturally produces two classes of excitations. Scalar chronons  $\delta\tau_n$  (spin 0) govern the dynamics of the condensate (mass, inflation), while vector modes  $A_\mu$  (spin 1), inherited from the 5D metric, are the geometric source of electromagnetism. This approach is fully consistent with the Kaluza–Klein theorem and removes the misunderstanding regarding the status of chronons as gauge bosons.

## 6. The Thermodynamic Arrow of Time as a Property of the Field

One of the deepest unresolved problems of fundamental physics is the origin of the arrow of time. The microscopic equations of motion (quantum mechanics, electrodynamics, gravity) are by nature symmetric under time reversal. Yet the macroscopic world is irreversible: entropy always increases, as captured by the Second Law of Thermodynamics. Traditional physics explains this purely statistically (through the probability of microstates), leaving open the question of why the universe began in an extraordinarily ordered, low-entropy state in the first place.

In the Temporal Theory of the Universe (TTU), the arrow of time ceases to be a statistical accident. It becomes a fundamental dynamical property of the time field  $\tau(x, \Theta)$  itself. In this section we show how the nonlinear dynamics of the temporal condensate physically generates macroscopic irreversibility.

### 6.1. The Potential $V(\tau)$ and Its Role in the Arrow of Time

As shown in Section 3, the dynamics of the temporal field is governed by a nonlinear potential  $V(\tau)$  of the spontaneous-symmetry-breaking type:

$$V(\tau) = (\lambda/4)(\tau^2 - \tau_0^2)^2, \quad (20)$$

where  $\tau_0$  is the vacuum expectation value of the field (the stable state) and  $\lambda$  is the self-coupling constant. In the early universe (during the Big Bang), the temporal field was in a high-energy state far from its true minimum  $\tau_0$ .

The “flow of time” that we experience is the process of global relaxation of the temporal field. It physically “rolls down” the slope of the potential  $V(\tau)$  toward the state of minimal energy. That is, the change in  $t_{\text{obs}}$  is a derivative of the change in  $\langle \tau \rangle$ . This process is fundamentally asymmetric: the directionality of time is the relaxation vector of the temporal condensate. The equation of motion obtained from this potential determines the dynamics:

$$\partial^2 \tau / \partial t_{\text{obs}}^2 + \gamma \partial \tau / \partial t_{\text{obs}} = -\partial V / \partial \tau, \quad (21)$$

where  $\gamma$  is the viscosity (dissipation) coefficient of the temporal medium. It is precisely this unidirectional gradient descent in the state space of the field  $\tau$  that sets the universal physical arrow of time.

### 6.2. Temporal Entropy: $S_\tau \propto \int [(\nabla \tau)^2 / \tau^2] dV$

If time is a field, it has its own entropy, measuring the degree of phase decoherence and spatial inhomogeneity. In TTU, the effective temporal entropy ( $S_\tau$ ) is defined through the integral of the squared relative gradient of the field:

$$S_\tau \propto \int [(\nabla \tau)^2 / \tau^2] dV, \quad (22)$$

where  $\nabla \tau$  is the spatial gradient of the time field.

- If the field is homogeneous and coherent ( $\nabla \tau = 0$ ), its temporal entropy is zero (the perfect order of the early universe).
- When matter appears (as a vortex configuration of the field) along with gravitational inhomogeneities, they create gradients  $\nabla \tau$ .

The greater the number of such inhomogeneities, the higher the temporal entropy becomes. This shows that entropy is not merely a count of states but a physical measure of the “disturbance” of the time field.

### 6.3. Relaxation of the Field and the Growth of Entropy

The growth of entropy thereby receives a clear mechanistic explanation. The interaction of matter (solitonic vortices of the field) inevitably leads to the dissipation of energy into the background temporal condensate.

Every physical process — the emission of light, the motion of planets, or chemical reactions — generates microscopic perturbations in the field  $\tau$ . These perturbations (emission of hypothetical quanta and phase decoherence) propagate through space, increasing the overall measure  $\langle(\nabla\tau)^2\rangle$ . The temporal condensate acts as a universal “heat bath,” absorbing information and losing its original global coherence. Relaxation of the field is inevitably accompanied by growth of microscopic disorder in its phase structure.

### 6.4. Connection to the Second Law of Thermodynamics

The Second Law of Thermodynamics ( $dS/dt \geq 0$ ) ceases to be an axiom in the TTU paradigm and becomes a theorem of the dynamics of the temporal field. The true law of thermodynamics is formulated for the complete system:

$$d(S_m + S_\tau)/dt_{\text{obs}} \geq 0. \quad (23)$$

Matter entropy ( $S_m$ ) always increases, because matter is inextricably linked to the time field  $\tau$ . Matter is “dragged along” by the global relaxation of the temporal medium, moving toward a state of maximal phase decoherence at the minimum of the potential  $V(\tau)$ . The heat death of the universe is the state in which the temporal field has reached its vacuum minimum ( $\tau = \tau_0$ ) and has become fully homogeneous ( $\nabla\tau = 0$ ).

**Conclusion to this section:** we finally lift the veil of mystery from the problem of irreversibility. Time has a direction not because that is how statistics happens to work, but because the fundamental field  $\tau(x, \Theta)$  is a dynamical physical system evolving toward its energy minimum. The Second Law of Thermodynamics is a direct consequence of the field equations governing the evolution of time.

## 7. Experimental Predictions and Observational Consequences

One of the key requirements for any fundamental physical theory is the existence of empirically testable consequences. If time is the primary quantum field  $\tau(x, \Theta)$ , and space, matter, and interactions arise as emergent manifestations of its dynamics, then this ontology must produce observable effects beyond the standard predictions. The predictions presented below should be regarded as directions for future experimental verification of the temporal paradigm.

## 7.1. Hypothetical Quanta of the Temporal Field: A Search in Cosmological Data

If the temporal field is quantized, its fundamental excitations — the scalar chronons  $\delta\tau_n$  (spin 0, see Section 5.3) — should manifest as discrete energy states. Just as the photon is the quantum of the electromagnetic field, the chronon is a localized excitation of the temporal condensate.

These quanta interact with ordinary matter extremely weakly (which makes them candidates for a “dark” sector), yet their presence may leave traces in the evolution of the early universe:

- **Effects on the cosmic microwave background (CMB):** chronons may contribute additional effective relativistic degrees of freedom ( $\Delta N_{\text{eff}}$ ), measurable by missions such as *Planck* or the *Simons Observatory*. Current constraints,  $N_{\text{eff}} \approx 2.99 \pm 0.17$ , allow an upper bound on  $m_\tau$  to be estimated.
- **Primordial fluctuations:** the fluctuation spectrum may contain nontrivial anomalies at small angular scales, caused by the quantum graininess of the temporal field during inflation.
- **Cosmological vacuum energy:** the nonlinear dynamics of the potential  $V(\tau) = (\lambda/4)(\tau^2 - \tau_0^2)^2$  may be a source of vacuum energy, explaining the cosmological constant as a residual effect of the field’s relaxation toward its minimum.

## 7.2. Deviations from the Law of Universal Gravitation at Micro-Scales

In the ontology of TTU, gravity is a structural-dynamical response of matter to the gradient of the logarithm of the temporal field (see Section 4.4:  $a = -\chi_\tau c^2 \nabla \ln \tau$ ). If scalar chronons have a finite mass  $m_\tau$ , this produces a Yukawa correction to the Newtonian potential:

$$V(r) = -(Gm/r)(1/\alpha) e^{-r/\lambda}, \quad \lambda \sim 1/m_\tau, \quad (24)$$

where  $\alpha$  is a dimensionless coupling constant determined by the parameter  $\chi_\tau$ . At sub-millimeter distances ( $r \lesssim 10^{-3}$  m), this correction may be detectable by high-precision torsion-balance experiments (of the Eöt-Wash type). Current constraints on the parameters  $\alpha, \lambda$  set a lower bound on the mass  $m_\tau \gtrsim 10^{-3}$  eV. If the compactification radius  $R_\Theta$  is at the Planck scale, then  $m_\tau \sim 10^{19}$  GeV, and the correction becomes unobservable at laboratory scales; however, if the LA-ODR connection ( $f_0 \approx 0.452$  MHz) points to a substantially smaller scale, this channel becomes critical for testing.

## 7.3. Anomalies in Signal Propagation Speed Under Strong Gradients

If the speed of light  $c$  is the effective propagation speed of perturbations in the equilibrium temporal condensate, then near sources of strong gravity, where the local field gradient  $\nabla\tau$  is extreme, a frequency-dependent dispersion should be observable.

### Experimental criteria:

- **High-energy photon delay:** high-energy gamma rays from distant gamma-ray bursts (GRBs) may “feel” the graininess of the field  $\tau$  more strongly than low-energy photons, leading to a frequency-dependent delay  $\Delta t \propto E_\gamma/M_{\text{LV}}$ , where  $M_{\text{LV}}$  is the Lorentz-invariance-violation scale associated with  $m_\tau$ . Current constraints from GRB 090510 give  $M_{\text{LV}} \gtrsim 10^{18}$  GeV.

- **Interferometric phase shifts:** in ultra-sensitive optical interferometers (of the LIGO/Virgo type, in astronomical mode) near massive bodies, phase anomalies not explained by the standard Shapiro effect may be observable if  $\nabla\tau \neq 0$ .

#### 7.4. Technological Applications: TPLL as a Resonant Phase Detector of Temporal Modes, and the Key Numerical Prediction of the LA-ODR Hypothesis

The concept of the **Temporal Phase-Locked Loop (TPLL)** involves building a highly coherent phase-locking system used not as an ordinary frequency stabilizer, but as an **active phase detector** of quantum excitations of the temporal condensate.

**Operating principle.** the TPLL compares the phase of an internal local oscillator,  $\phi_{LO}$ , with the phase of fluctuations of the temporal field,  $\phi_\tau$ . The output signal of the phase detector is proportional to the sine of the phase difference:

$$V_{out} \propto \sin(\Delta\phi), \quad \Delta\phi = \phi_{LO} - \phi_\tau. \quad (25)$$

When the local-oscillator frequency  $\omega_{LO}$  is swept near the resonance frequency  $\omega_n$  of the quantized temporal-field modes, a characteristic **lock-in effect** (phase capture) should appear, manifesting as:

- a sharp jump in the output voltage  $V_{out}$  upon crossing resonance;
- a characteristic modulation of the system's phase-noise spectrum, correlated with  $\omega_n$ .

The resonance condition is determined by the dispersion relation for scalar chronons (Section 5.2):

$$\omega_{LO} = \omega_n = \sqrt{[\kappa(n/R_\Theta)^2 + (m_\tau c/\hbar)^2]}, \quad n = 1, 2, 3, \dots \quad (26)$$

Thus the TPLL becomes a direct instrument for registering individual temporal modes: every jump or anomaly in the phase noise at a given frequency  $\omega_n$  would be evidence of the excitation of the corresponding chronon mode.

**Key numerical prediction of the LA-ODR hypothesis.** according to the LA-ODR hypothesis (see Section 7.1 of [7]), the dimensionless parameter  $R_{\Theta,LA} \approx 1.245$  is identified with the physical compactification radius  $R_\Theta$ . Under this identification, for the fundamental mode  $n = 1$  (with  $m_\tau \ll 1/R_\Theta$ ), one obtains the frequency:

$$f_0 = \omega_1/2\pi = c/(2\pi R_\Theta) \approx 0.452 \text{ MHz}, \quad \lambda_0 \approx 663 \text{ m}. \quad (27)$$

If this signal is detected as a lock-in effect in ultra-high-coherence experiments, it would constitute strong support for the LA-ODR connection. It is important to stress that this prediction belongs to the LA-ODR hypothesis, not to TTU itself; TTU provides only the geometric framework within which such an identification is possible, but not mandatory. The passage from the dimensionless parameter  $R_{\Theta,LA}$  to the physical frequency of 0.452 MHz is an additional postulate of the hypothesis, not a consequence of the Kaluza–Klein reduction itself (Sections 4.1–4.2).

**Falsification criterion.** the absence of a characteristic phase jump or noise anomaly at the frequency  $f_0 \approx 0.452 \text{ MHz}$  (Eq. 27) in a controlled experiment would constitute a direct refutation of the LA-ODR hypothesis. It is important to emphasize that this would not refute the Kaluza–Klein reduction itself (Theorem 1, Sections 4.1–4.2), which is mathematically independent; it would only show that the identification of  $R_\Theta$  with  $R_{\Theta,LA}$  was incorrect. The baseline compactification scale corresponding to the KK reduction without additional hypotheses remains

at the Planck scale (Section 7.1), and the connection to the laboratory frequency of 0.452 MHz remains, in that case, entirely unsupported.

In addition, the TPLL allows for:

- **Detection of local anomalies:** registration of shifts in the lock-in frequency correlated with the position of massive objects or with the Earth’s daily rotation (indicating a change in the gradient  $\nabla\tau$  and the local effective mass  $m_\tau$ ).

## 7.5. Methodological Limitations and Open Problems

This work does not attempt a derivation of spacetime from absolute nothingness — a separate and substantially harder program pursued elsewhere in quantum cosmology. The claim of TTU is narrower: given the minimal background manifold  $M_5$ , the physical content built upon it — metric, curvature, matter, and the gravitational and gauge interactions — is shown to emerge from the dynamics of a single temporal field, rather than being introduced as independent postulates. Several aspects of this construction remain open and are stated explicitly below, so that the reader can distinguish established results from working hypotheses.

(i) *The background manifold.* The manifold  $M_5 = M_4 \times S^1$  — including its dimensionality, signature, and the compactness of  $\Theta$  — is postulated as an effective scaffold on which the field  $\tau(x, \Theta)$  is defined; it is not derived from  $\tau$  in this work. What is derived here is the effective metric, curvature, and gravitational dynamics induced by  $\tau$  on this fixed background (Sections 4.1–4.4, Appendix D). A deeper, pre-geometric derivation of  $M_5$  itself — for example from the phase structure or correlation geometry of the temporal condensate — remains an open problem for future work.

(ii) *The thermal interpretation of  $\Theta$ .* As noted in Section 3.2, the identification of  $\Theta$  with a Euclidean thermal circle is an interpretive analogy rather than a proven equivalence with finite-temperature field theory.

(iii) *The three-generation bound.* The saturation argument of Section 4.3 does not predict exactly three generations from first principles with no free input; rather, it shows that the geometric saturation criterion  $|\nabla\tau| < \tau_{\text{crit}}$  provides a plausible mechanism for the observed three-generation pattern. A dedicated numerical study of the parameter dependence of  $n_{\text{max}}$  is left for future work.

(iv) *The LA-ODR frequency scale.* The derivation of the dimensionless infrared fixed point  $R_{\Theta, \text{LA}} \approx 1.245$  lies outside the scope of the present paper and is given in Ref. [7]; here it is used only as an input to the falsifiable prediction of Section 7.4.

(v) *Terminology.* The term *chronon* has an established historical usage denoting a discrete, minimal unit of time (Caldirola, 1980; Snyder, 1947). In the present work it is repurposed to denote a scalar excitation of the temporal field  $\tau$  (Section 5.3) and should not be conflated with that earlier literature.

## 8. Conclusions: Why This Step Is Inevitable for a Complete Theory

Revising the fundamental nature of time is one of the most difficult conceptual steps in science. Historically, physics has already made comparable radical shifts: the transition from absolute space to relativistic spacetime, and from determinism to quantum probability. The Temporal Theory of

the Universe (TTU) maintains that the next logical stage in the development of fundamental physics is to abandon time as a passive coordinate and to grant it the status of a primary quantum field. This step is not a philosophical indulgence — it is dictated by the internal crises of theoretical physics itself.

### 8.1. Compliance with the Principles of Physics: Logical, Mathematically Consistent, Falsifiable

Any new paradigm must satisfy strict criteria of scientific methodology. The proposed ontology of the temporal field fully meets these requirements:

- **Logical necessity:** the theory does not add new entities for the sake of complexity; on the contrary, it applies the principle of emergence. It explains otherwise mysterious phenomena (the unidirectionality of time, the origin of space and mass) through a single universal mechanism.
- **Mathematical consistency:** this paper demonstrates that the concept of a time field requires no exotic or untested mathematics. It relies on canonical quantization, differential geometry, and multidimensional (Kaluza–Klein) reduction. The effective Maxwell action,  $-1/4 F_{\mu\nu} F^{\mu\nu}$ , is reproduced through the standard Kaluza–Klein reduction (Sections 4.1–4.2), and the metric and Ricci curvature are expressed as functionals of the field  $\tau$ , demonstrating the emergent character of geometry within the adopted approximations.
- **Falsifiability (Popper’s criterion):** the theory generates concrete, testable predictions (Section 7). The existence of hypothetical field quanta (chronons), the modification of the gravitational potential at small scales, and effects in highly coherent phase-locked systems make it open to experimental confirmation or refutation.

### 8.2. Summary of the Four Arguments

To unify quantum mechanics and general relativity into a single theory of quantum gravity, the transition to a field ontology of time is architecturally inevitable. This conclusion rests on four pillars, examined in detail in this paper:

- **Resolution of the problem of time.** the fundamental carrier of dynamics is the quantum temporal field  $\tau$ , while the macroscopic parameter  $t$  is merely its vacuum expectation value:  $t \sim \langle \tau \rangle$ . This ontological distinction removes the “frozen universe” paradox (the Wheeler–DeWitt equation), since the field  $\tau$  possesses its own conjugate momentum and canonical algebra, giving the system an internal referent of evolution. The dynamics of quantum gravity is restored as a natural consequence of the dynamics of the temporal condensate, without need for an external clock.
- **Occam’s razor (ontological reduction):** space, matter (topological vortices), and gauge interactions (phase transport) reduce to a single common denominator — the states and gradients of the temporal condensate. The arena and the actors merge into a single entity.
- **Mechanism for the regularization of singularities:** owing to the compact hyper-temporal dimension  $\Theta$ , the field is inevitably quantized. The existence of a discrete spectrum opens a natural mechanism for regularizing the singularities of black holes and the Big Bang, one that requires further numerical verification within specific collapse models.

- **The dynamical arrow of time:** the thermodynamic asymmetry of the universe ceases to be a purely statistical accident and becomes a direct consequence of the relaxation of the temporal field along its nonlinear potential. The growth of entropy is microscopic noise arising from the global macroscopic expansion and settling of time.

### 8.3. A Theorems vs. Hypotheses Summary Table

To avoid conflating rigorous theorems with physical hypotheses, we reduce all of the key claims of this paper to two categories. This allows the reader to clearly distinguish results that are irrefutable consequences of the mathematical formalism from those that remain open questions or interpretations requiring further experimental verification.

| What Is Proven Mathematically (Theorems)                                                                                                                                          | What Is Interpretation / Hypothesis (Open Questions)                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quantization of the field $\tau$ on the compact dimension $S^1$ gives a discrete spectrum of modes $E_n = \hbar\sqrt{[k(n/R_\Theta)^2 + (m_\tau c/\hbar)^2]}$ (Sections 5.1–5.2). | The energy spectrum may provide regularization of black hole and Big Bang singularities (proof that quantum effects dominate over classical collapse precisely at the Planck scale is still required; this is at present an open problem).                                                                                                                                                                                                   |
| Kaluza–Klein reduction of the 5D metric on $M_4 \times S^1$ yields the 4D Maxwell action $-1/4 F_{\mu\nu}F^{\mu\nu}$ (Sections 4.1–4.2, refs. [1, 2]).                            | Scalar chronons $\delta\tau_n$ are gauge bosons (in fact, only the vector modes $g_{\mu\Theta}$ are gauge fields; the scalar modes $\delta\tau$ are dilatons — the distinction is given in Section 5.3).                                                                                                                                                                                                                                     |
| The potential $V(\tau) = (\lambda/4)(\tau^2 - \tau_0^2)^2$ has a global minimum at $\tau = \tau_0$ , ensuring a stable vacuum (Section 6.1).                                      | The “flow of time” is relaxation toward this minimum (the thermodynamic arrow) — this is a dynamical interpretation that requires a numerical relation between the relaxation rate and the observed age of the universe.                                                                                                                                                                                                                     |
| The conformal metric $g_{\mu\nu} = (\tau/\tau_0)\eta_{\mu\nu}$ gives the scalar curvature $R[\tau] = (\tau_0/\tau)[-3\Box\tau/\tau + (3/2)(\partial\tau/\tau)^2]$ (Section 4.2).  | A possible geometric explanation of the three-fermion-generation bound is based on the temporal-gradient saturation criterion ( $ \nabla\tau  < \tau_{\text{crit}}$ ), without introducing free gauge parameters. Within the proposed model, the upper bound $n_{\text{max}} \approx 3$ is determined exclusively by the geometric parameters of the medium. Final numerical verification of this result remains a task for future research. |
| The equation of motion $\Box_5\tau + \partial V/\partial\tau = 0$ is derived from a variational principle (Section 3.4).                                                          | The source of the initial excitation of $\tau$ in the early universe (the problem of initial conditions: why the field started in a high-energy state rather than directly in the vacuum).                                                                                                                                                                                                                                                   |
| Definition of the temporal entropy $S_\tau \propto \int [(\nabla\tau)^2/\tau^2] dV$ (Section 6.2).                                                                                | The LA-ODR connection between the compactification radius $R_\Theta$ and the frequency                                                                                                                                                                                                                                                                                                                                                       |

|                                                                                                       |
|-------------------------------------------------------------------------------------------------------|
| $f_0 \approx 0.452$ MHz is a hypothesis that requires direct experimental verification (Section 7.4). |
|-------------------------------------------------------------------------------------------------------|

## 8.4. Outlook: From Theory to Engineering

Recognizing that time is a quantum field fundamentally changes our place in the universe. As long as time was regarded as an abstract coordinate, we could only be its passive observers (passengers). However, any physical field can be not only measured but also locally perturbed.

If electromagnetism began with the abstract Maxwell equations and culminated in radio physics and electronics, then recognizing the field-theoretic nature of time opens the way to an entirely new discipline — temporal engineering. Instruments for direct temporal metrology (such as the TPLL systems discussed here) could become the first devices to register fluctuations of this medium. In the longer term, control of the phase coherence of temporal modes might allow local influence over the effective geometry of spacetime and the inertial properties of matter.

**Conclusion to this section:** we have shown that the transition from time-as-coordinate to time-as-field  $\tau(x, \Theta)$  makes it possible to:

- eliminate the Wheeler–DeWitt paradox;
- reduce space and matter to a single temporal condensate;
- propose a natural mechanism for the regularization of singularities through quantization on  $S^1$ , one that requires numerical verification;
- explain the arrow of time as relaxation of the potential  $V(\tau)$ .

The proposed model outlines concrete experimental criteria, in particular the LA-ODR hypothesis ( $f_0 \approx 0.452$  MHz), which may be verified in experiments using TPLL systems. This makes TTU not merely a mathematical construction but a physical program open to falsification, in which every numerical prediction is clearly identified by its origin.

**Overall summary:** turning time into a dynamical field is not merely a change of mathematical symbols. It is a fundamental paradigm shift that transforms time from a background clock into an active physical substance. Only by recognizing the materiality of time can we hope to build a truly complete, unified theory of the universe.

## Appendix D. Step-by-Step Derivation of the Scalar Curvature of the Conformal Metric (Supplement to Section 4.2)

The conformal metric of this paper,  $g_{\mu\nu} = (\tau/\tau_0)\eta_{\mu\nu}$ , is a special case of the standard conformal transformation  $g_{\mu\nu} = e^{2u}\eta_{\mu\nu}$  with exponent

$$u = (1/2) \ln(\tau/\tau_0). \quad (D1)$$

The standard transformation formula for the scalar curvature under a conformal transformation of a flat metric in  $n$  dimensions (Wald, General Relativity, Eq. D.10; Carroll, Spacetime and Geometry, Eq. 3.46) reads:

$$R = e^{-2u} [-2(n-1)\Box u - (n-1)(n-2)(\partial u)^2], \quad (D2)$$

where  $\Box$  and  $\partial$  are evaluated with respect to the flat (background) metric  $\eta$ . For  $n = 4$  (ordinary four-dimensional spacetime):

$$R = -6e^{-2u} [\Box u + (\partial u)^2]. \quad (D3)$$

Substituting  $u = (1/2) \ln(\tau/\tau_0)$  and using the standard identity for the Laplacian of a logarithm,  $\Box \ln f = \Box f/f - (\partial f/f)^2$ , we obtain:

$$\Box u = (1/2)[\Box \tau/\tau - (\partial \tau/\tau)^2], \quad (\partial u)^2 = (1/4)(\partial \tau/\tau)^2. \quad (D4)$$

Substituting into the bracket:

$$\Box u + (\partial u)^2 = (1/2)\Box \tau/\tau - (1/2)(\partial \tau/\tau)^2 + (1/4)(\partial \tau/\tau)^2 = (1/2)\Box \tau/\tau - (1/4)(\partial \tau/\tau)^2. \quad (D5)$$

Multiplying by  $-6$  and using  $e^{-2u} = \tau_0/\tau$ :

$$R[\tau] = (\tau_0/\tau)[-3\Box \tau/\tau + (3/2)(\partial \tau/\tau)^2]. \quad (D6)$$

This confirms the coefficients  $-3$  and  $3/2$  given in Section 4.2, and it explicitly fixes the conformal prefactor  $\tau_0/\tau$ , which is easily dropped when the formula is written in abbreviated form. Note that the derivatives  $\Box$  and  $\partial$  in the final formula are evaluated with respect to the flat background metric  $\eta$ , not the emergent metric  $g$ ; this is the standard convention when working with conformal transformations and must be stated explicitly in any subsequent use of this result.

## References

- [1] Th. Kaluza, *Zum Unitätsproblem der Physik*, Sitzungsber. Preuss. Akad. Wiss. Berlin **1921**, 966–972.
- [2] O. Klein, *Quantentheorie und fünfdimensionale Relativitätstheorie*, Z. Phys. **37** (1926) 895–906.
- [3] J. M. Overduin and P. S. Wesson, *Kaluza–Klein Gravity*, Phys. Rept. **283** (1997) 303–380. (A review providing a detailed account of Kaluza–Klein reduction and its applications.)
- [4] D. Bailin and A. Love, *Kaluza–Klein Theories*, Rept. Prog. Phys. **50** (1987) 1087–1170. (An additional reference on modern versions of KK theories.)
- [5] P. S. Wesson, *Space–Time–Matter: Modern Kaluza–Klein Theory*, World Scientific, Singapore, 1989. (A monograph developing the “induced matter” approach.)
- [6] J. I. Kapusta and C. Gale, *Finite-Temperature Field Theory: Principles and Applications*, 2nd ed., Cambridge University Press, 2006. (Used for the analogy with the thermal circle  $S^1_\beta$ .)
- [7] A. Lemeshko and I. Ohnishi, *LA-ODR Mechanism for an Infrared Leptonic Resonance Scale*, TTU-Series Preprint, 2026. (Source of the hypothesis relating  $R\Theta$  to the infrared fixed point and the frequency  $f_0 \approx 0.452$  MHz.)
- [8] P. Caldirola, *The introduction of the chronon in the electron theory and a charged-lepton mass formula*, Lett. Nuovo Cimento **27** (1980) 225–228.

- [9] H. S. Snyder, *Quantized Space-Time*, Phys. Rev. **71** (1947) 38–41.